Comparative Risk Detection Across Banking Problems: Fraud, Money Laundering and Customer Churn

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Abstract

Banks are facing several distinct risks, including fraudulent transactions, money laundering and customer attrition, yet each is usually studied in isolation with its own model and evaluation protocol. This paper compares Decision Tree, Random Forest and XGBoost classifiers on three publicly available banking datasets: PaySim (fraud, 6.36M transactions), SAML-D (money laundering, 9.50M transactions) and a Botswana bank customer churn dataset (115,640 customers) using one shared experimental framework. The framework combines dataset-specific feature engineering, leakage control, a 70/30 train test split, SMOTE applied only inside the training folds of a stratified five-fold cross-validation, and model selection by PR-AUC followed by F1-score. The selected models are then evaluated once on an untouched test set. Random Forest was selected for PaySim (test F1 = 99.98 %, PR-AUC = 99.99 %), and XGBoost for SAML-D (F1 = 82.38 %, PR-AUC = 93.59 %) and churn (F1 = 97.71 %, PR-AUC = 99.86 %). The SAML-D result shows a clear precision recall trade-off (precision 74.13 %, recall 92.70 %) that overall accuracy (99.96 %) completely hides. The results indicate that no single model is guaranteed to be best across banking risks: the most suitable choice depends on the dataset, the nature of the risk, and the balance between missed cases and false alarms.

Index Terms fraud detection, anti-money laundering, customer churn, class imbalance, SMOTE, XGBoost, Random Forest, precision recall.

I. Introduction

Banks operate under several kinds of risk at once. Fraudulent transactions cause direct financial loss, money laundering exposes institutions to regulatory and reputational damage, and customer churn erodes long-term revenue. Machine learning is widely used to flag all three, but the problems differ sharply in what a "positive" case looks like, how rare it is, and what a wrong prediction costs. A missed fraud or laundering case is expensive, while a false alarm consumes analyst time; in churn prediction, in contrast, a false alarm often only costs a retention offer.

A common difficulty is *class imbalance*. In the two transaction datasets studied here, fewer than 0.13 % of records are positive, so a classifier that labels everything as legitimate would be more than 99.8 % accurate while detecting nothing. Accuracy is therefore a poor yardstick, and the precision recall trade-off is what matters in practice [1, 2].

Most published work treats fraud, anti-money-laundering (AML) and churn separately [3, 4, 5], which makes it hard to say whether conclusions about a model carry over from one risk to another. This study addresses that gap with a controlled comparison. The same three algorithms, the same imbalance-handling strategy, the same validation scheme and the same metrics are applied to three banking problems, so that differences in performance can be attributed mainly to the data and the risk rather than to the experimental setup.

Our research question is *"How effectively can machine-learning models detect different types of banking risk, and how does their performance vary across fraud detection, money laundering detection and customer churn prediction?"*. In this paper, we test three models (Decision Tree, Random Forest and XGBoost) on three banking problems: fraud, money laundering and customer churn. We use the same steps for all three so the results are fair to compare. We choose the best model using PR-AUC and F1-score instead of accuracy, because the data is very imbalanced. We also look at how long training takes, what false alarms and missed cases mean in practice, and where our results should be treated with care.

II. Background and Related Work

A. Risk Detection in Banking

Fraud detection, AML monitoring and churn prediction are all supervised binary classification problems on tabular data, but they differ in structure. Fraud detection is a transaction-level problem in which signals such as inconsistent balance changes or unusual transaction types matter. AML monitoring is concerned with patterns of movement across accounts, and is typically evaluated at the transaction level when labels are available. Churn prediction operates at the customer level and relies on tenure, product holdings, balances and service history.

B. Class Imbalance and Resampling

Learning from imbalanced data is a well-studied problem [6]. The Synthetic Minority Over-sampling Technique (SMOTE) creates new minority examples by interpolating between a minority sample and its nearest minority neighbours [7]. Resampling must be applied only to training data; if it is applied before splitting or before cross-validation, synthetic points derived from validation records can leak into training and inflate scores. In this study SMOTE is therefore placed inside the cross-validation pipeline.

C. Tree-Based Classifiers

Decision trees [8] are interpretable but high-variance. Random Forests [9] average many decorrelated trees and are robust on tabular data. Gradient-boosted trees, of which XGBoost [10] is a widely used implementation, build trees

sequentially to correct earlier errors and often achieve the best accuracy on structured data. These three models span the range from an interpretable baseline to a strong ensemble.

D. Evaluation Under Imbalance

Because the precision recall curve focuses on the positive class, it is generally more informative than the ROC curve on heavily imbalanced data [2, 1]. We therefore use the area under the precision recall curve (PR-AUC), computed as average precision, as the primary selection metric, with F1-score as the tie-breaker, and report ROC-AUC, precision, recall and accuracy for completeness.

E. Datasets in the Literature

PaySim is a simulator of mobile-money transactions built from aggregated real logs [11]. SAML-D is a synthetic transaction-monitoring dataset created to support AML research, with labelled laundering typologies [12]. The Botswana bank customer churn dataset is publicly distributed through Kaggle [13]. Using these three together allows a side-by-side comparison across risk types.

F. Related Work

One study compared Decision Tree, Random Forest and XGBoost on PaySim using random undersampling. Random Forest was the most accurate (about 99%), followed by XGBoost (98.9%) and Decision Tree (97%) [3]. Another study oversampled PaySim and compared Logistic Regression, Gradient Boosting and XGBoost, where XGBoost performed best (accuracy 99.88%) and threshold moving was used to tune detection [14]. A third study combined a Local Outlier Factor stage with LightGBM and SMOTE on PaySim [15]. On a credit-card fraud dataset, Random Forest and XGBoost with SMOTE outperformed the other classifiers, and the authors noted that the age of the data limits generalization [16].

For money laundering, a transformer-based model tested on SAML-D reached an ROC-AUC of 93.01 with a 98% true-positive rate and a 51% false-positive rate, and it outperformed TabTransformer and XGBoost [4]. An interview study with eight AML specialists found that rule-based transaction monitoring produces many false positives, and it suggested graph analysis and anomaly detection as improvements [17].

For churn, a comparison of Random Forest and XGBoost on a bank dataset with SMOTE and SMOTE-ENN found that XGBoost with SMOTE was best (F1 $= 0.873$, AUC $= 0.983$) [5]. Another study varied the churn rate from 15% to 1% in a telecom setting. XGBoost with SMOTE gave the highest F1-score at every imbalance level, while Random Forest performed poorly under severe imbalance [18].

Each of these studies covers a single risk domain, and some rely on random splits [3, 16]. Our work applies one leakage-aware protocol to all three domains, with time-ordered splits for the transaction data, so the domains can be compared directly.

III. Datasets

Table I summarizes the three datasets, and Fig. 1 compares their positive-class rates. PaySim and SAML-D are extremely imbalanced (0.129 % and 0.104 % positive), whereas churn is moderately imbalanced (12.19 %).

Table I
DATASET OVERVIEW

	PaySim	**SAML-D**	**Churn**
Risk problem	Fraud	Laundering	Churn
Records	6,362,620	9,504,852	115,640
Positive cases	8,213	9,873	14,094
Positive rate	0.129%	0.104%	12.19%
Train records	4,453,834	6,653,396	80,948
Test records	1,908,786	2,851,456	34,692
Train positives	3,643	6,764	9,866
Test positives	4,570	3,109	4,228
Train / test rate	0.082 / 0.239%	0.102 / 0.109%	12.19 / 12.19%
Split type	Chronological	Chronological	Stratified

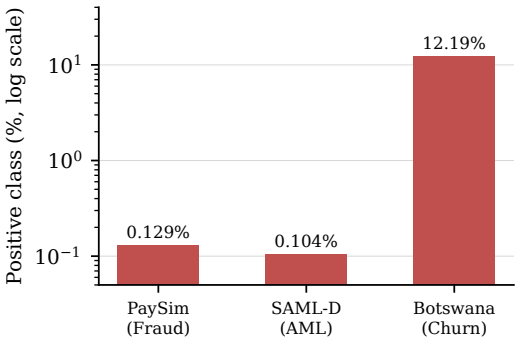


Fig. 1. Positive-class rate in each dataset (log scale). Fraud and laundering cases are roughly two orders of magnitude rarer than churn cases.

Each record is a mobile-money transaction with a time step (hour), transaction type, amount, origin and destination balances before and after the transaction, and a fraud label. The original `isFlaggedFraud` field was removed, as it is a rule-based flag generated by the simulator rather than an input feature. No missing values or duplicates were found.

Each record is a transaction with date, time, sender and receiver accounts, amount, payment and received currency, sender and receiver bank locations, payment type, a laundering label, and a laundering-typology label. The typology label was excluded, since it only exists for laundering cases and would leak the target.

Each record is a customer with demographic attributes, income, tenure, credit information, outstanding loans, balance, number of products and complaints, and a churn flag. `Churn Reason` and `Churn Date` are populated only for churned customers (101,546 missing values, equal to the number of retained customers) and were removed to prevent target leakage.

IV. Methodology

Fig. 2 shows the pipeline applied to every dataset. The same sequence of steps is used throughout; only the feature engineering and a small number of model settings are dataset-specific.

A. Feature Engineering

Features were designed per dataset to capture the signals most relevant to each risk (Table II). Identifiers (account

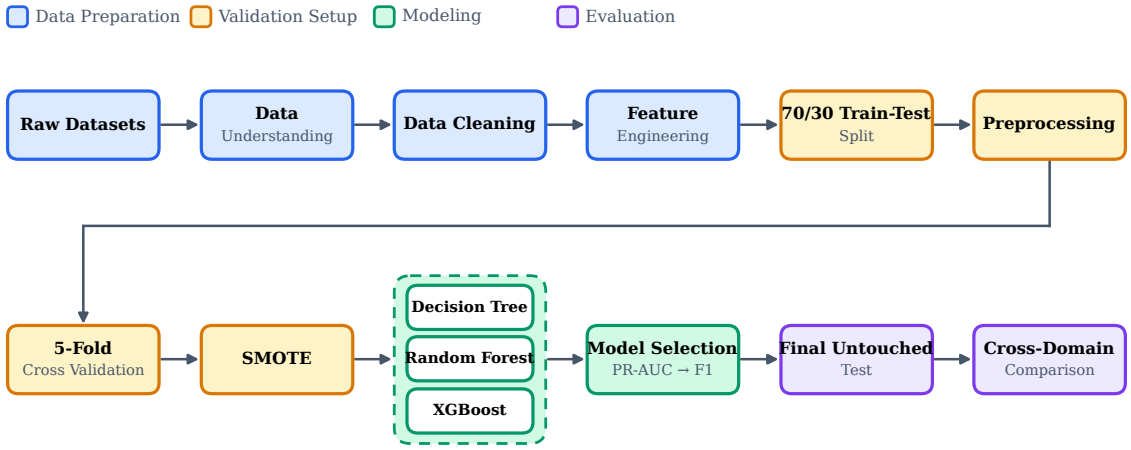


Fig. 2. Overview of the methodology pipeline shared by all three banking problems, from raw data through preprocessing, SMOTE-based cross-validation, model selection and the final untouched test.

names, customer IDs, names, addresses and contact details) were removed.

In PaySim and SAML-D the running counts are computed with a cumulative count over the chronologically sorted data, so each record only sees earlier activity of the same account and no information flows backwards from the future.

B. Train Test Split

The two transaction datasets are sorted by time and split chronologically, with the first 70% used for training and the last 30% for testing. This mimics deployment, where a model trained on the past scores future transactions. The churn dataset has no temporal ordering, so a stratified random 70/30 split (seed 42) was used, which preserves the 12.19% churn rate in both partitions.

C. Preprocessing and Imbalance Handling

Categorical variables are one-hot encoded (unseen categories ignored). For PaySim and churn, numerical variables are standardized; for SAML-D they are passed through unchanged. SMOTE is then applied to the *training portion of each fold only*, using an `imblearn` pipeline in which preprocessing, SMOTE and the classifier are fitted together, so validation folds are never resampled. The sampling ratio (minority-to-majority after resampling) was 0.1 for PaySim and SAML-D, which are extremely imbalanced, and 0.5 for churn. For PaySim, SMOTE used $k = 5$ neighbours.

D. Models

Three classifiers are compared in every domain: a Decision Tree (interpretable baseline), a Random Forest (bagged ensemble) and XGBoost (gradient boosting). Their hyperparameters (Table III) were fixed in advance and not tuned per dataset beyond adjusting tree depth and number of trees to dataset size. All use random seed 42.

E. Validation and Model Selection

On each training set, stratified five-fold cross-validation (shuffled, seed 42) was run for all three models with SMOTE inside each fold. The model with the highest mean PR-AUC was selected, with F1-score used as the secondary criterion. The selected model was then retrained on the entire training set (with SMOTE) and evaluated exactly once on the held-out test set.

F. Evaluation Metrics

With true positives TP, false positives FP, false negatives FN and true negatives TN:

$$\text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN}, \tag{1}$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{2}$$

Precision measures how many raised alerts are correct, and recall measures how many true cases are found. PR-AUC summarizes the precision recall trade-off across all thresholds, and ROC-AUC summarizes the true-positive versus false-positive trade-off. All thresholds for hard predictions are the default 0.5 on predicted probability; no threshold tuning was performed.

V. Results

A. Cross-Validation Comparison

Table IV reports the five-fold cross-validation results for all three models, and Fig. 3 shows PR-AUC by model. The selected model in each domain is marked in bold.

On PaySim the top two models are very close: Random Forest has a mean PR-AUC of 0.9948 versus 0.9943 for XGBoost, a gap of 0.0005. Random Forest was selected because the protocol ranks on PR-AUC first, even though XGBoost had a higher CV F1-score (0.9930 vs. 0.9862) and higher precision, and ran about nine times faster (736 s vs. 6,929 s). On SAML-D XGBoost had the best PR-AUC, ROC-AUC and F1, while Random Forest had the highest precision and the Decision Tree the highest recall. On churn XGBoost led on every metric, while Random Forest was clearly the weakest ($F_1 = 0.827$).

B. Final Test Performance

Table V gives the held-out test results for the selected models, and Fig. 4 compares precision, recall and F1.

PaySim and churn are detected almost perfectly. SAML-D is clearly harder: recall is high (92.70 %), but precision

Table II
ENGINEERED FEATURES AND LEAKAGE CONTROLS PER DATASET

Dataset	Engineered features	Removed (leakage / identifiers)
PaySim	Origin and destination balance errors (old − amount − new and old + amount − new); zero-balance flags; log transforms of amount and balances; amount-to-balance ratios; running transaction counts for sender and receiver; same-account flag; transfer/cash-out flag. (1 categorical + 21 numerical features.)	Sender and receiver names, `isFlaggedFraud`.
SAML-D	Currency mismatch; bank-location mismatch and same-location flag; log amount and zero-amount flag; hour, day of week and weekend flag; running counts for sender, receiver and sender receiver pair. (5 categorical + 12 numerical features.)	`Laundering_type`, sender and receiver accounts, raw date and time.
Churn	Age (from date of birth); balance-to-income and loan-to-income ratios; log of balance, income and loans; products per tenure; balance per product; complaints per tenure. (6 categorical + 18 numerical features.)	`Churn Reason`, `Churn Date`, IDs, names, address, contact information, date of birth.

Table III
MODEL HYPERPARAMETERS

Model	Setting	PaySim	SAML-D	Churn
Decision Tree	max depth	12	none	10
	min leaf samples	10	2	10
Random Forest	trees	100	200	200
	max depth	20	none	20
	min leaf samples	5	2	5
XGBoost	trees	200	300	300
	max depth	6	6	6
	learning rate	0.10	0.10	0.05
	subsample / colsample	0.8 / 0.8	0.8 / 0.8	0.8 / 0.8
SMOTE	sampling ratio	0.1	0.1	0.5

is only 74.13 %, so about one in four alerts is a false alarm. Note that SAML-D accuracy is 99.96 %, higher than churn accuracy (99.44 %), even though SAML-D is by far the weaker detector. This is a direct illustration of why accuracy is misleading under severe imbalance.

C. Error Analysis

Fig. 5 shows the confusion matrices. On PaySim only one legitimate transaction was flagged and one fraud missed among 1,908,786 test records. On SAML-D, 2,882 of 3,109 laundering transactions were detected, 227 were missed and 1,006 legitimate transactions were flagged. On churn, 4,163 of 4,228 churners were identified, with 65 missed and 130 false alarms.

D. Cross Validation Versus Test

Fig. 6 compares cross-validated and test scores. Churn is stable (PR-AUC 0.9982 → 0.9986, F1 0.9751 → 0.9771). PaySim improves on the test set (PR-AUC 0.9948 → 0.9999). SAML-D shows the largest shift: PR-AUC rises from 0.827 to 0.936, while F1 slightly falls (0.840 → 0.824) and the balance of precision and recall changes completely (CV precision 0.982 and recall 0.734, test precision 0.741 and recall 0.927). This pattern is discussed in Section VI.

VI. DISCUSSION

Random Forest was selected for fraud, and XGBoost for money laundering and churn. The PaySim margin was very small, so the stronger claim is not that Random Forest is better for fraud but that the ranking of models was not the same in all three domains and that the size of the gap between them varies widely (large on churn, small on PaySim). This

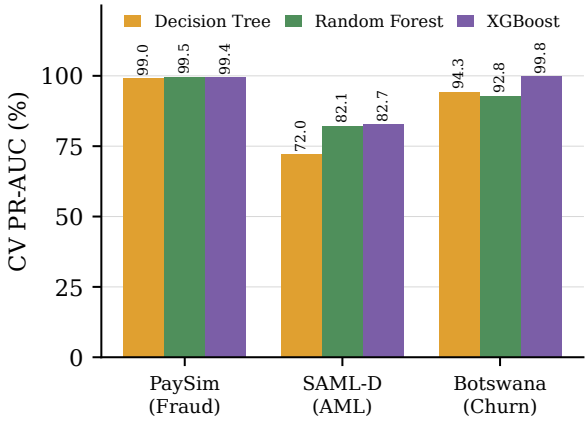


Fig. 3. Cross-validated PR-AUC of the three classifiers in each domain. XGBoost leads on SAML-D and churn; on PaySim, Random Forest and XGBoost are almost tied.

supports the central message that model choice should be made per problem.

Our churn results agree with recent work in which XGBoost with SMOTE was the strongest model and Random Forest degraded under imbalance [5, 18]: in our CV, XGBoost led on churn while Random Forest was weakest. The PaySim ordering (Random Forest ≥ XGBoost > Decision Tree on PR-AUC) matches the ordering reported in [3], although their protocol (undersampling, random split) differs from ours. The Tab-AML result on SAML-D [4] is not directly comparable to ours, because the split, features and operating point differ; it does, however, point to the same difficulty we observe, namely a trade-off between detecting laundering and generating false alerts.

PaySim's engineered balance-error features expose a strong, almost deterministic signature of fraud in the simulated data, which likely explains the near-perfect test scores; whether real-world fraud is as separable is not something this experiment can show. Churn benefits from informative customer-behaviour features and a moderate 12.19 % positive rate. SAML-D is the hardest: laundering is defined by patterns across accounts and time, while the features used here are mostly transaction-level, so many laundering transactions look ordinary in isolation. Richer network or time-window features are a natural next step.

On SAML-D, 1,006 false alerts accompany 2,882 true detections. For an AML team this is roughly 0.35 false alert

Table IV

FIVE-FOLD CROSS-VALIDATION RESULTS (SMOTE INSIDE FOLDS). MEAN OVER FOLDS; SELECTED MODEL IN BOLD.

Dataset	Model	Acc.	Prec.	Rec.	F1	ROC-AUC	PR-AUC	CV time (s)
PaySim	Decision Tree	0.99990	0.8952	0.9920	0.9407	0.9980	0.9897	571
	Random Forest	**0.99998**	**0.9810**	**0.9915**	**0.9862**	**0.9991**	**0.9948**	**6,929**
	XGBoost	0.99999	0.9950	0.9909	0.9930	0.9992	0.9943	736
SAML-D	Decision Tree	0.99968	0.8965	0.7725	0.8298	0.8925	0.7205	448
	Random Forest	0.99965	0.9885	0.6634	0.7938	0.9939	0.8212	5,840
	XGBoost	**0.99972**	**0.9824**	**0.7343**	**0.8404**	**0.9955**	**0.8267**	**2,464**
Churn	Decision Tree	0.9804	0.9033	0.9400	0.9212	0.9841	0.9428	39
	Random Forest	0.9545	0.7708	0.8923	0.8270	0.9863	0.9283	117
	XGBoost	**0.9939**	**0.9682**	**0.9822**	**0.9751**	**0.9997**	**0.9982**	**155**

Table V

FINAL HELD-OUT TEST PERFORMANCE OF THE SELECTED MODEL IN EACH DOMAIN (%)

Dataset	Model	Acc.	Prec.	Rec.	F1	ROC-AUC	PR-AUC	Train (s)	Test (s)
PaySim (Fraud)	Random Forest	99.9999	99.9781	99.9781	99.9781	100.0000	99.9999	1,812.9	19.6
SAML-D (AML)	XGBoost	99.9568	74.1255	92.6986	82.3782	99.9475	93.5894	592.6	43.1
Churn	XGBoost	99.4379	96.9718	98.4626	97.7115	99.9803	99.8634	43.0	0.6

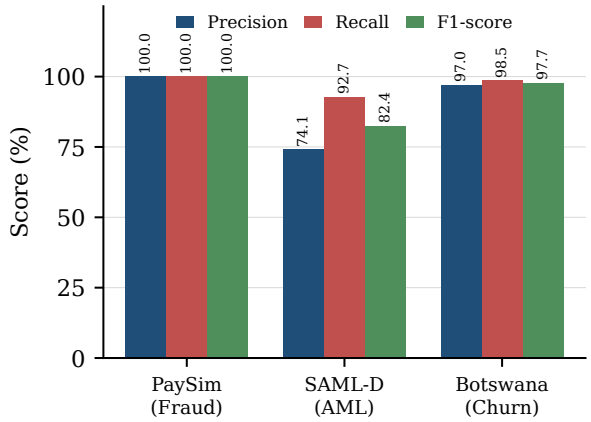


Fig. 4. Test precision, recall and F1-score of the selected model in each domain. Precision is the weak point on SAML-D.

per correct one, which may be acceptable if missing laundering is much costlier than reviewing an alert, but it must be a deliberate choice. A decision threshold other than 0.5 could trade recall for precision; this was not explored. On churn, the cost of a false positive (an unnecessary retention offer) is low, so the high recall is appropriate.

In cross-validation SAML-D had high precision (0.98) and modest recall (0.73); on the test set the pattern reversed. We have not isolated the cause. Two plausible contributors are that cross-validation shuffles chronologically ordered training data while the test set is a later time period, and that the laundering mix may differ between periods. This is a reason to treat the SAML-D CV numbers as a guide to model ranking rather than as an estimate of test performance.

Accuracy exceeded 99.4% in all three domains, but F1 ranged from 82.4% to 99.98%. PR-AUC, precision and recall expose differences that accuracy hides, supporting the use of PR-AUC for selection.

VII. LIMITATIONS

1. **Synthetic or simulated data.** PaySim and SAML-D are simulation-based, so performance may not transfer to

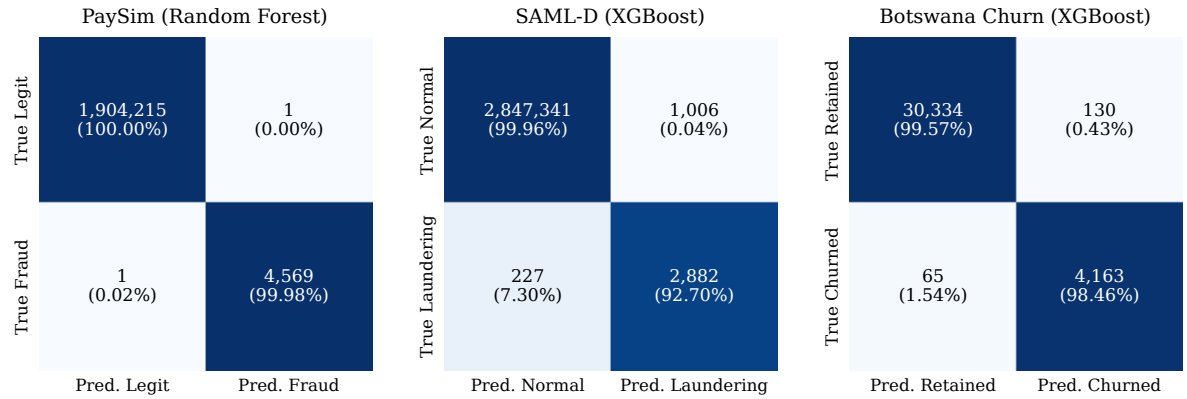


Fig. 5. Confusion matrices on the held-out test sets (counts and row-normalized percentages).

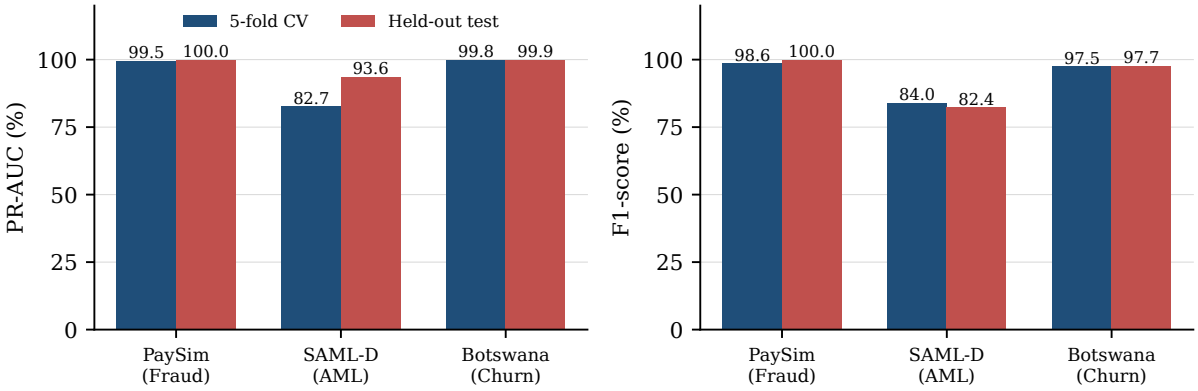


Fig. 6. Cross-validation versus held-out test PR-AUC (left) and F1-score (right).

production systems.

2. **Separate datasets for the three problems.** We could not find a single transaction dataset that contains fraud, money laundering and customer churn labels together. The three problems therefore come from three different datasets with different sources, sizes, features and time periods. Differences in performance reflect both the type of risk and the dataset itself, so the cross-domain comparison is not perfectly controlled.

VIII. Future Work

Future work should first address the two limitations above. The models should be tested on real banking data, to check whether the results hold outside simulated datasets. A better design would also use one dataset, or linked datasets from the same bank, that contains fraud, money laundering and churn labels together, so the three problems can be compared under identical conditions. Other useful steps are to tune the hyperparameters and report results over several random seeds, to use time-aware cross-validation for the transaction data, to add graph-based and time-window features for money laundering, to tune decision thresholds using the real cost of false alarms and missed cases, to examine feature importance, and to compare SMOTE with class weighting and other resampling methods.

IX. Conclusion

This paper compared Decision Tree, Random Forest and XGBoost for fraud, money laundering and churn detection under one leakage-aware framework with SMOTE inside cross-validation and PR-AUC-based model selection. Random Forest was selected for PaySim and XGBoost for SAML-D and churn. Fraud and churn were detected with F1-scores of 99.98 % and 97.71 %, while money laundering remained harder (F1 82.38 %, precision 74.13 %, recall 92.70 %). Accuracy was above 99.4 % in all cases and so could not distinguish the problems. The main conclusion is that performance and the best model are problem-specific: no model was universally best, and the right choice depends on the data, the risk, and the balance between missed cases and false alarms. The results should be interpreted within the stated limitations, in particular the simulated nature of two datasets and the use of three separate datasets, one for each problem.

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